Water Quality & Potability Prediction



Logistic Regression Binary Classification

AI101 (2026) · IIT Ropar | Instructors: Dr. Puneet Kumar & Dr. Rakesh Sanodiya

GROUP 77 TEAM

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GitHub: github.com/yatharth-2026aib1099/AI101-Group77-WaterPotability

Problem & Dataset

Goal: given 9 water properties, predict whether water is safe to drink (1 = Potable) or unsafe (0 = Not Potable).

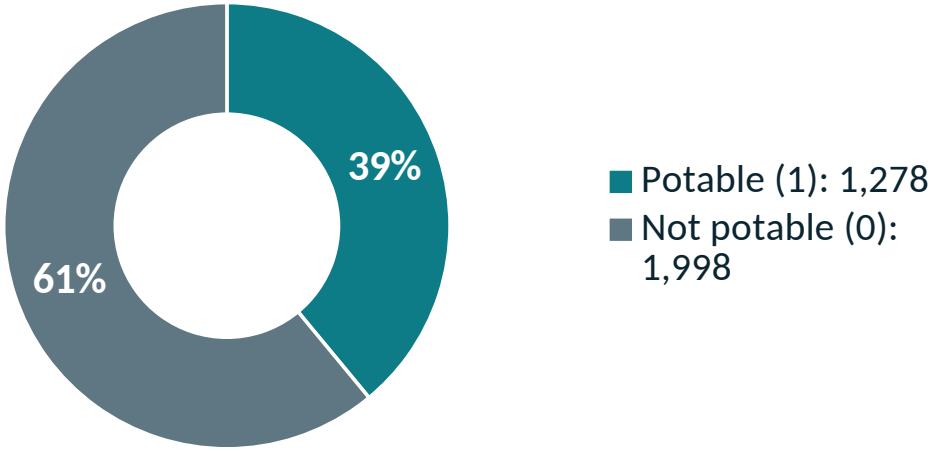
3,276
samples (Kaggle Water Quality Dataset)

9
water-property features

Features

- pH
- Hardness
- Solids
- Chloramines
- Sulfate
- Conductivity
- Organic Carbon
- Trihalomethanes
- Turbidity

Target classes



Missing values: only 3 columns

491	781	162
pH	Sulfate	Trihalomethanes

Pipeline (How We Built It)

1

Handling Missing Values

- Replaced missing values with the median of each column
- pandas .fillna()

2

Train-Test Split (80/20)

- Train: 2,620 samples
- Test: 656 samples
- stratify=y keeps the same 0/1 ratio

3

Feature Scaling (StandardScaler)

- Mean = 0, Variance = 1
- Solids is ~20,000 while pH is only ~7

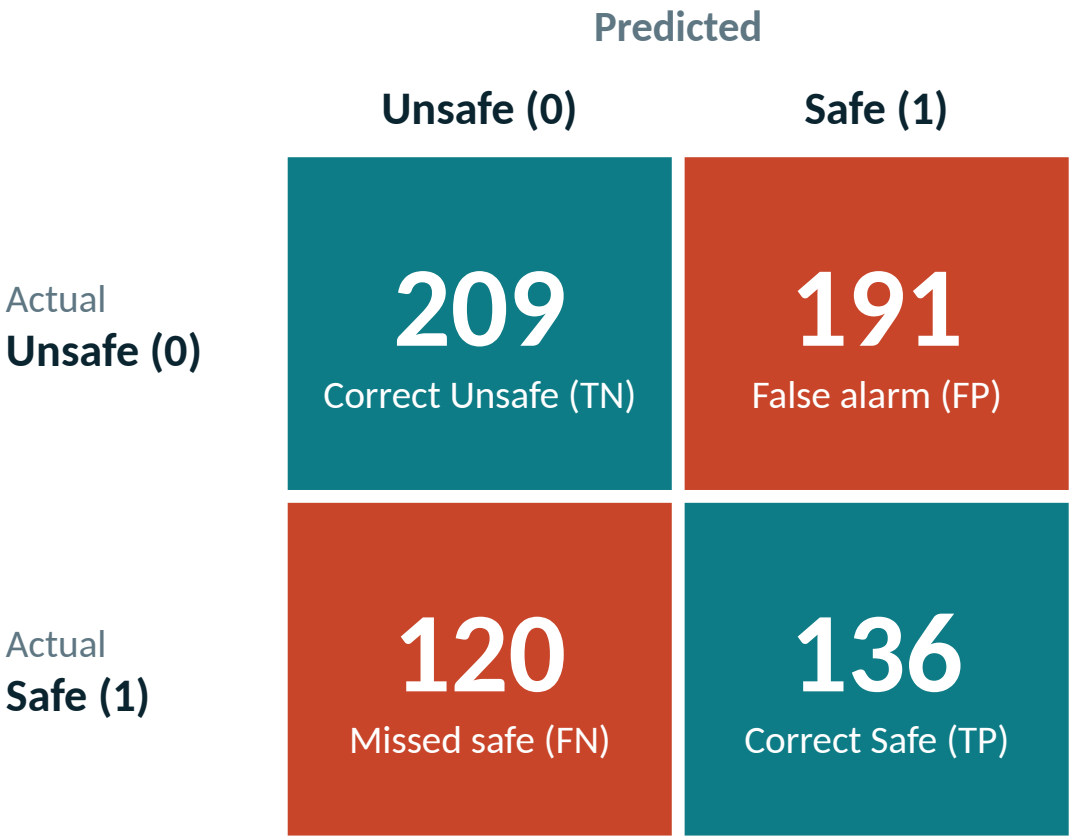
4

Logistic Regression

- Standard scikit-learn logistic model
- Outputs 0 or 1

Results & Confusion Matrix

Confusion matrix (656 test samples)



~53% Accuracy	**41.6%** Precision
53.1% Recall	**0.47** F1-score

Recall: catches more than half of all potable water samples.

What influences the model most?

Higher **Solids, Chloramines** and **pH** increase the chance of safe water.

Higher **Organic Carbon** decreases the chance of safe water.

Key Learnings & AI Disclosure

Project Insights

- Logistic Regression is fast, simple, and easy to interpret.
- Real-world water chemistry is complex; future work could test non-linear models like Random Forest.

Team Contribution

- Every single member contributed working Python code committed from their own GitHub account.

AI Disclosure (Rubric Compliance)

- ChatGPT was used for research.
- Claude was used for slide preparation.
- All code was adapted from our lab practice, tested, and run locally by Group 77 leads.